Electric Fields

Exam code: 0625 & 0972

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Electric fields

Extended tier only

- An electric field is defined as:

A region of space in which an electric charge experiences a force

- The direction of an electric field at a point is defined as:

The direction of the force on a positive charge at that point

- Charged objects **create** electric fields around themselves
  - This is similar to how magnets create magnetic fields
- An electric field is a **vector** quantity as it has both **magnitude** (strength) and **direction**

Electric field patterns

Extended tier only

- Electric field lines are
  - used to represent the **direction** and **magnitude** of an electric field
  - always directed from the **positive** charge **to** the **negative** charge

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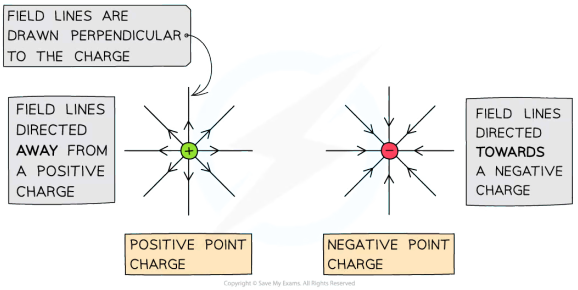
Got it

radially inwards or outwards:
  - If the charge is **positive** (+), the field lines are radially **outwards**
  - If the charge is **negative** (-), the field lines are radially **inwards**

Electric field lines around positive and negative point charges

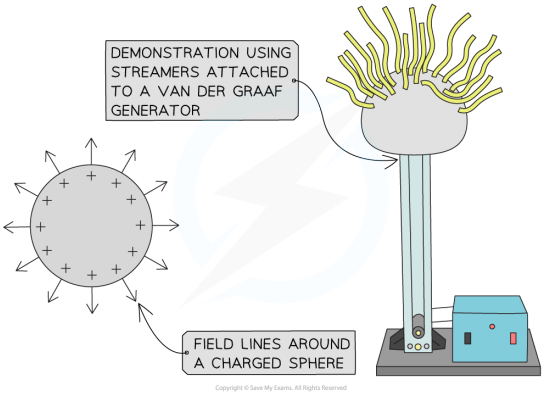


Electric field lines around a point charge are directed away from a positive charge and towards a negative charge

Electric field around a charged conducting sphere

- The field lines around a charged conducting sphere are similar to that of a point charge
  - This is because the charges on the surface of the sphere will be evenly distributed
  - The charges are the same, so they repel
  - The surface is conducting, allowing them to move
- Field lines are **always perpendicular** (at right angles) to the surface of a conducting sphere

Electric field lines around a positively charged sphere

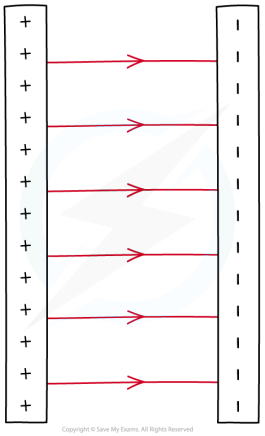


The electric field pattern around a conducting can be demonstrated using a Van der Graaff generator

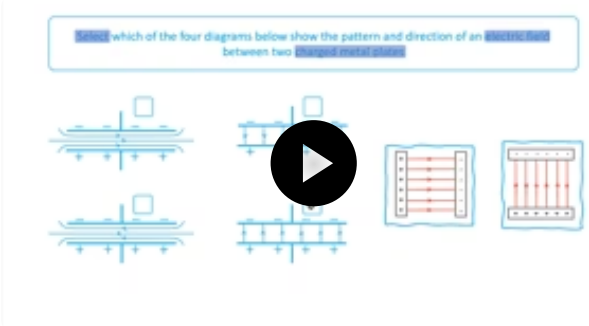
Electric field between two parallel plates

- directed from the **positive** to the **negative** plate
- parallel
- straight lines

Electric field lines between two oppositely charged parallel plates



Electric field lines between two parallel plates are directed from the positive to the negative plate. A uniform electric field has equally spaced field lines



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